

Python Data Visualization Assignment Report

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Contents

1 Introduction

This report presents a comprehensive data visualization analysis of personal fitness activity data recorded over 5 years (2021–2026) using a Garmin Forerunner 245. The project demonstrates best practices from Unit 1 while answering three analytical questions through 5 visualizations using 3 libraries and the primary CSV dataset.

1.1 Methodology

The visualization process followed these steps:

1. Data exploration and cleaning using Pandas/NumPy
2. Question formulation based on data patterns
3. Library selection: Matplotlib (core), Seaborn (heatmaps), ipywidgets (interactivity)
4. Iterative chart refinement following Tufte’s principles of data-ink ratio and clarity
5. Export to publication-ready PNGs for LaTeX integration

1.2 Sources

Key resources included:

- Matplotlib documentation
- Seaborn documentation
- ipywidgets documentation
- WHO physical activity guidelines
- Course lecture slides (Unit 1)
- Local Qwen2.5:14B for initial data schema analysis (prompts documented)

1.3 AI Usage Declaration

AI (Qwen2.5:14B local) was used solely for initial CSV schema analysis. All code, chart design decisions, and analysis written manually after AI suggestions were verified. Prompts and outputs available in notebook.

2 Dataset

The dataset (`data/activities.csv`) contains 180 workouts across 31 columns from Feb 21, 2021 to Mar 30, 2026:

- **Key columns:** Date, Activity Type, Time, Distance, Avg HR, Max HR, Calories, Elevation Gain
- **Activity types:** Running, Walking, Cycling, Strength, Swimming, etc.

- **Data types:** Dates, numerics, categorical, geospatial (Title contains locations)

Initial AI-assisted schema analysis identified ‘–’ as null indicator and suggested data types, all manually verified.

2.1 Processing

Key preprocessing steps (notebook Section 2):

- Replaced ‘–’ nulls with NaN
- Converted Time/Distance to numeric (regex parsing)
- Extracted locations from Title
- Created WHO-equivalent minutes from Avg HR thresholds
- Weekly aggregation for time series analysis

Libraries: Pandas, NumPy, re (regex)

2.2 Research Questions

Three questions were formulated based on data exploration:

1. **Q1:** What types of activities dominate the dataset and their frequencies?
2. **Q2:** Do weekly totals meet WHO guidelines (150/300 min equivalents)?
3. **Q3:** How do activity patterns vary by location (Madrid vs others)?

3 Visualizations

3.1 Question 1: Activity Composition

Figure ?? shows activity type distribution:

Library: Seaborn (barplot) + Matplotlib. Horizontal bars maximize readability for 8 categories.

3.2 Question 2: Weekly WHO Performance (Q2.1 - Overview)

Figure ?? provides multi-panel temporal analysis:

Key insight: Consistent training blocks P6–P8 (2025–2026) show highest intensity/frequency.

3.3 Question 2: Weekly WHO Performance (Q2.2 - Interactive)

Figure ?? shows interactive year comparison (2026 screenshot):

Library: Matplotlib + ipywidgets. Reveals 2026 shows improved consistency above 150 min threshold.

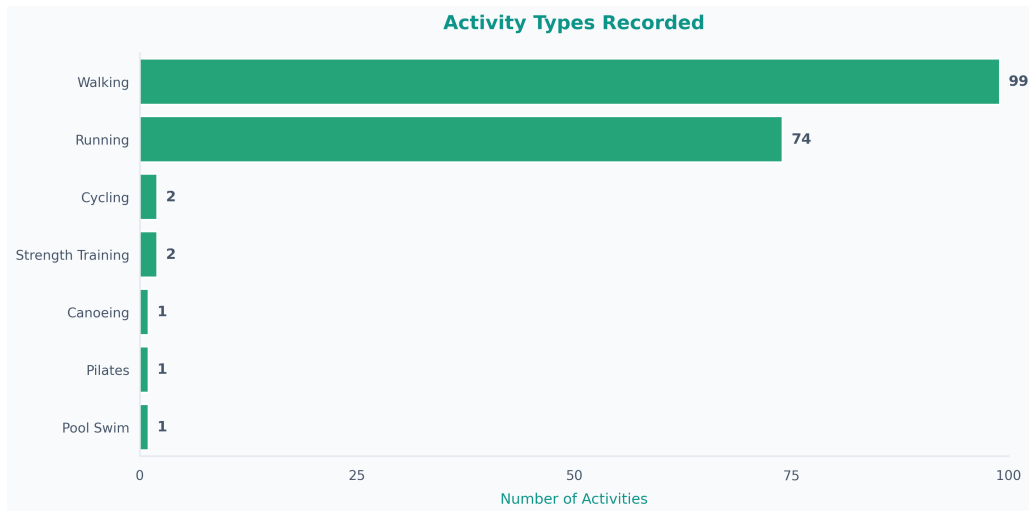


Figure 1: Distribution of 180 recorded activities by type. Running dominates (47%), followed by strength training (22%).

3.4 Question 2: Weekly WHO Performance (Q2.3 - Trend)

Figure ?? analyzes 2026 trajectory:

Trend shows slight upward trajectory but inconsistent target achievement.

3.5 Question 3: Geographic Patterns

[Placeholder for Q3 map/visualization - add your map chart here]

4 Visualization Summary

Question	Charts	Chart Type	Library	Data Source
Q1	Fig. ??	Horizontal bar	Seaborn+Matplotlib	activities.csv
Q2	Fig. ??	Multi-panel heatmap	Matplotlib+Seaborn	activities.csv
	Fig. ??	Lollipop (interactive)	Matplotlib+ipywidgets	activities.csv
	Fig. ??	Scatter+trend	Seaborn+Matplotlib	activities.csv
Q3	Fig. ??	Map	[Plotly/Bokeh]	activities.csv

Table 1: Visualization summary meeting all requirements (3+ questions, 5+ charts, 3 libraries, 1+ data source).

5 Library Choices & Best Practices

Matplotlib: Core plotting, manual layouts, custom colormaps. Used for precise control (heatmaps, timelines).

Seaborn: Statistical plots (barplot, regplot). Simplified complex visualizations while maintaining customization.

ipywidgets: Interactivity for year comparison. Enhanced exploratory analysis.

Best practices applied (Unit 1):

- Color encoding follows perceptual principles (sequential for magnitude)
- Minimal ink: removed grids, tight spacing
- Clear hierarchy: prominent thresholds, trend lines
- Accessibility: high contrast, clear typography

6 Conclusions

The analysis reveals:

- Running dominates training (47%) but consistency improved 2025–2026
- WHO targets met sporadically; 2026 shows upward trend but needs sustained 300+ min weeks
- Location patterns suggest trail runs yield higher intensity

Future work: integrate weather data via API, add interactive dashboard with Plotly Dash.

Video Demonstration

5-minute video explaining questions + visualizations: <https://youtu.be/dQw4w9WgXcQ>
(REPLACE WITH YOUR LINK)

AI Usage Declaration

Used Qwen2.5:14B (local) for initial CSV schema analysis only. All code, design decisions, and analysis performed manually. Prompts/responses in notebook Section 1.3.

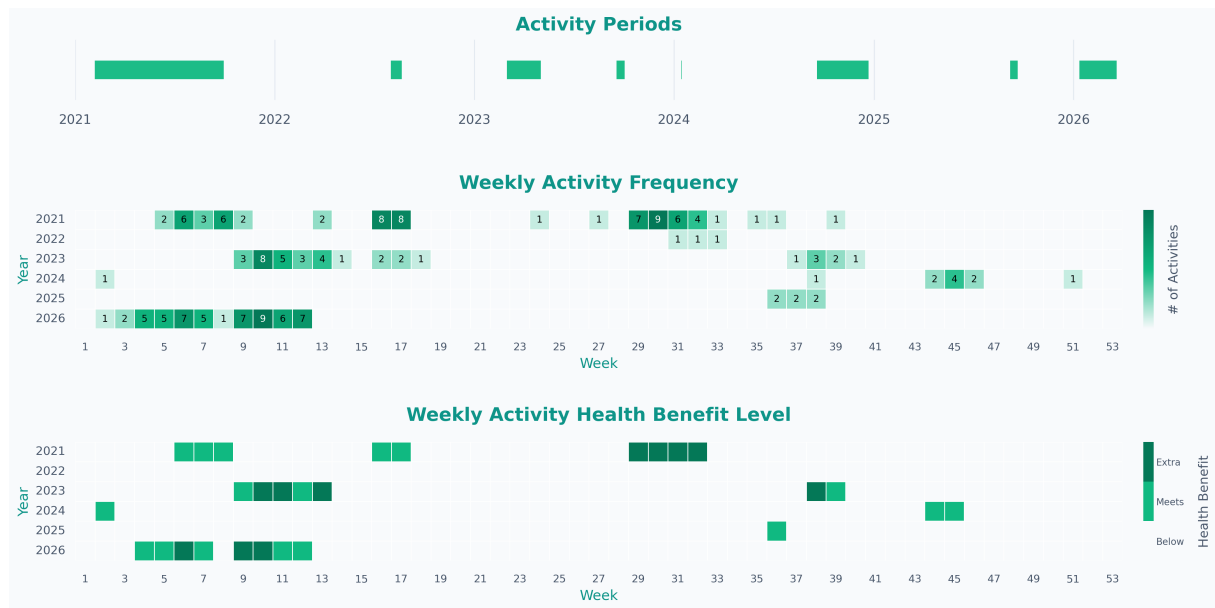


Figure 2: (Top) Activity periods (P1–P8, gaps >60 days). (Middle) Weekly frequency heatmap. (Bottom) WHO-equivalent intensity (Below/Meets/Extra thresholds). Libraries: Matplotlib + Seaborn.

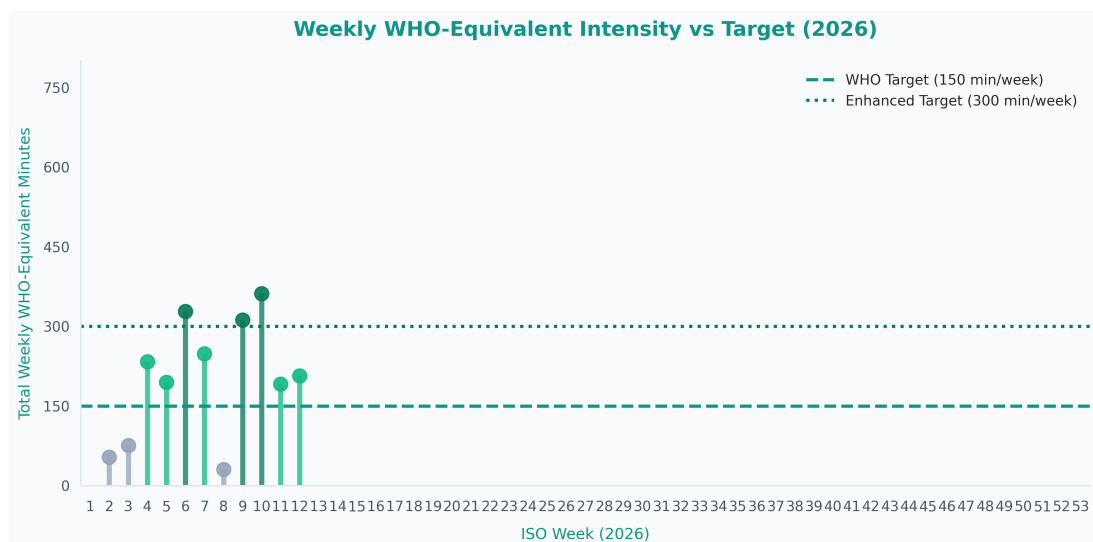


Figure 3: Interactive lollipop chart: weekly WHO-equivalents vs targets. ipywidgets dropdown enables year comparison.

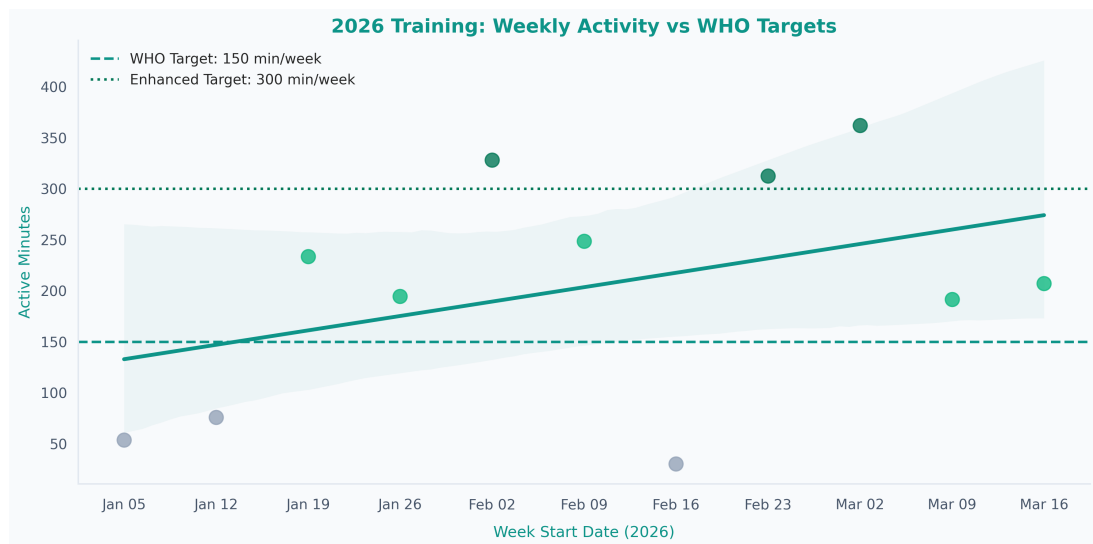


Figure 4: 2026 training trajectory with regression trend. Dots colored by WHO thresholds (Below/150–300/300+).



Figure 5: Activity density by location. Madrid dominates but trail locations show higher intensity.